

PLATOLIVES

Version 3.5 User and Technical Manual
A native PLATO terminal for modern macOS

REAL COLOR CRT (CDC 721 / IST-III) | CALIBRATED OPTICAL TUBE CURVATURE
P22 PHOSPHOR PERSISTENCE | ZERO IDLE CPU

512 x 512 logical display | 2048 x 2048 analog rendering | Native Cocoa, Core Animation & Metal-aligned BGRA
macOS 13+ | Apple Silicon M1-M4 and modern Intel Macs

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About This Manual

This manual describes PlatoLives 3.5 from two complementary perspectives. The first part is intended for users who connect to Cyber1 or alternative PLATO services, manage connection profiles, operate multiple simultaneous terminal windows, use the PLATO keyboard and touch model, select monochrome or color display modes, calibrate optical distortion and persistence, collect diagnostic information, or automate repeatable sessions. The final part documents the C11 terminal architecture and the 60 FPS optical presentation pipeline in sufficient detail to support maintenance, calibration, verification, and future platform ports.

Scope: The document reflects the PlatoLives 3.5 design and the consolidated September 2026 source snapshot. Features identified as future work are intentionally not presented as available.

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1. Why PlatoLives Exists

1.1 The practical motivation

Pterm is an important and feature-rich terminal emulator that has enabled the PLATO community to access Cyber1 and related systems for many years. PlatoLives is a small, lightweight, fully native alternative tailored to current macOS systems, Apple Silicon, and modern Intel hardware. It complements the work that came before it. Version 3.5 extends preservation beyond protocol fidelity into calibrated optical fidelity: authentic monochrome plasma, full-color CDC 721 / IST-III streams, physically moderate tube curvature, P22 phosphor decay, and true static-screen quiescence.

1.2 Project goals

- Run natively on macOS 13 or later on Apple Silicon M1 through M4 and modern Intel Macs.
- Compile the portable libplato core as pure C11 with -Wall -Wextra -Wpedantic -O2.
- Use Cocoa/AppKit, Core Animation, layer-backed views, and Metal-aligned native BGRA memory without obsolete X11 or GTK dependencies.
- Support independent multi-window, multi-host sessions with profile-specific optical calibration.
- Preserve the 512 x 512 PLATO logical contract while rendering analog simulations on a continuous 2048 x 2048 surface.
- Reach deterministic 60 FPS during active animation and exactly 0.0% idle CPU after persistence energy has fully decayed.
- Keep protocol, fonts, framebuffer operations, persistence, distortion, input inversion, and presentation independently testable.

1.3 Preservation without stagnation

The protocol, terminal behavior, character sets, graphics modes, touch semantics, and coordinate system are the historical contract. Networking, memory layout, optical reconstruction, GPU presentation, diagnostics, and user-interface management are modern engineering choices.

Design position: Historical fidelity belongs in observable terminal behavior. Modern engineering belongs in the implementation.

2. Design Principles

Principle	Meaning in PlatoLives 3.5
Speed	60 FPS active presentation, O(1) separable blurs, tile-bounded recomputation, and zero idle CPU.
Clean foundation	Native Cocoa/AppKit shell and pure C11 core without Rosetta, X11, or legacy GUI frameworks.
Portability	Protocol, framebuffer, graphics primitives, canonical fonts, color state, and persistence live in reentrant libplato.
Determinism	The 512 x 512 logical terminal is authoritative; optical rendering is a decoupled 2048 x 2048 presentation stage.
Physical restraint	Curvature is calibrated to $k = 0.018$ rather than exaggerated into unreadable fisheye projection.
Verification	Protocol regression tests, inspectable transforms, inverse touch mapping, and non-breaking menu/profile migration.
Visibility	Mach CPU telemetry, diagnostic stream logging, dirty-tile counters, and command-file automation.

3. Getting Started and Connection Profiles

3.1 Launch behavior

PlatoLives 3.5 loads the designated default connection profile, preconfigured for Cyber1 on TCP port 8005, and opens an independent session window. The profile applies its saved full-screen state, display mode, plasma settings, CRT settings, and applicable distortion mode. The 512 x 512 PLATO surface remains a central 1:1 square. On 4:3 CRT simulations it is pillarboxed inside the physical tube model; pixels outside the optically warped boundary are pitch black.

3.2 Connecting to Cyber1

- Launch PlatoLives and wait for the IST-III negotiation and sign-on screen.
- Press Return, mapped to PLATO NEXT.
- At User, enter demo1 and press Return.
- At Group, enter demo and press Shift+F10 or Shift+Ctrl+S for SHIFT-STOP.
- At Password, press Return for the public demonstration account. Personal accounts should enter the assigned password and press NEXT.

Connection target: The supplied default profile targets cyberserv.org on TCP port 8005.

3.3 Dynamic Connection Profiles (Cmd+,)

Each profile is both a transport definition and an optical presentation preset. Plasma controls are used only when Real Plasma is active. CRT controls are used only when Real Color CRT is active. Crisp and Split Monochrome suppress analog controls to prevent irrelevant or contradictory configuration.

+-----+ Connection Profiles +-----+			
+-----+ Profile Settings			
Cyber1 *	Profile Name:	[Cyber1	]
IRATA.ONLINE	Server Host:	[cyberserv.org	]
Local Test	Port:	[8005]	
	Display Mode:	[Real Color CRT	v]
	Plasma Persist:	[100 ms	v]
	Plasma Distort:	[Cylindrical	v]
[+] [-]	CRT Beam:	[High (Soft Glow)	v]
+-----+	CRT Persist:	[20 ms (P22 Fast)	v]
	CRT Distort:	[Cylindrical	v]
	[x] Launch Full Screen		
	[x] Default at startup	[Save] [Connect]	
+-----+ +-----+ +-----+			

Profile rules

- Create, duplicate, edit, and delete profiles with the + and - controls.
- Connect Now closes the active socket, clears logical and analog buffers, resets protocol state, applies the selected profile, and reconnects immediately.
- The window shows Plasma Persistence and Plasma Distortion only for full-screen Real Plasma.
- The window shows CRT Beam, CRT Persistence, and CRT Distortion only for Real Color CRT.
- Crisp Monochrome, Split Monochrome, and Crisp Color hide analog calibration controls.
- Connection and status-bar menus synchronize immediately through NSMenuDelegate.

3.4 Multi-Window Sessions and the Status Bar Companion

Cmd+N opens an independent window using the default profile. New Window with Profile is available from both File and the status-bar menu. Each window owns an isolated socket, network worker, terminal RAM, logical color matrix, persistence buffers, dirty tiles, timer state, and CALayer. ESC X metadata updates the title using the form PlatoLives: <user>/<group> (<slot>) - <ProfileName> (<host>:<port>). The NSStatusItem menu reports current state and offers fast access to profiles, reconnect, disconnect, About, and Quit.

4. How PlatoLives Works

4.1 Network and protocol lifecycle

An asynchronous POSIX network `worker` receives stream bytes with `TCP_NODELAY` active and writes into a thread-safe 64 KB circular ring buffer. A display-side scheduler drains bounded chunks through the PLATO state machine. The initial `0x0D` probe awakens compatible ASCII services and begins the ESC Y terminal identification exchange. On disconnect the worker returns to a low-cost listening state, enabling unlimited reconnect cycles without accumulating threads or sockets.

4.2 Dual logical framebuffers

The historical plane remains a 512 x 512 one-bit bitboard:

```
term->fb.pixels[512][64]
```

Version 3.5 adds a parallel full-color matrix carrying one native little-endian BGRA word per logical cell:

```
term->fb.colors[512][512]
```

Monochrome operations preserve the bitboard as the compatibility authority. Color-capable drawing updates both semantic occupancy and logical color. The optical stage reads these structures and reconstructs the selected 2048 x 2048 output without changing protocol coordinates.

4.3 Cyber1 Real Color protocol

During hardware identification PlatoLives responds to the subtype-16 color query with ESC Y `0x71`. This enables the Cyber1 full 32-bit color stream associated with CDC 721 / IST-III behavior. Incoming color control selects the active foreground and background values, and subsequent points, vectors, blocks, and glyphs write the selected BGRA color into the logical matrix. Canonical geometric INVERSE, mode = 3, writes `bg_color` rather than toggling an abstract bit, including block fills. This resolves filled prompt anomalies while preserving host-selected color semantics.

4.4 Coordinate system and toroidal arithmetic

Logical coordinate (0,0) is at the bottom-left. Memory rows are top-down, therefore:

```
raster_y = 511 - y
```

Backspace and Step use true modulo-512 arithmetic. For any horizontal displacement `d`:

```
x' = (x + d) mod 512
```

In C, negative results are normalized as $((x + d) \% 512 + 512) \% 512$. This makes the display a torus in the horizontal direction and prevents clipping at the left edge, including the historical halved-knight failure in Camelot.

4.5 Canonical typography and EXT scaling

M0 and M1 remain the canonical CERL 8 x 16 sets. M2 and M3 are programmable RAM fonts encoded as 18-bit words and loaded through ESC P and ESC S. Version 3.5 implements continuous CDC Appendix G EXT character scaling for octal codes 005100 through 005177:

```
scale = (size_val + 7) / 16
```

The scaler applies the same continuous factor to glyph rasterization, tracking, and cursor advance. Multi-size sequences therefore retain proportion and do not collapse or overlap, including very large title graphics such as the Asteroids heading.

5. Keyboard and Input Reference

PlatoLives provides a unified keyboard engine supporting historical Pterm conventions and modern macOS shortcuts.

PLATO key	Primary macOS shortcut	Alternate / technical note
NEXT	Return or Enter	Ctrl+N, Option+N
SHIFT-NEXT	Shift+Return	Shift+Ctrl+N
BACK	Esc or F8	Ctrl+B, Option+B
SHIFT-BACK	Shift+Esc or Shift+F8	Shift+Ctrl+B
STOP	F10 or F4	Ctrl+S, Option+S, Cmd+S
SHIFT-STOP	Shift+F10 or Shift+F4	Shift+Ctrl+S, Shift+Option+S
ERASE	Backspace or Delete	Left Arrow, Ctrl+R, Option+R
HELP	F6 or F1	Ctrl+H, Option+H
LAB	F7 or F2	Ctrl+L, Option+L
DATA	F9 or F3	Ctrl+D, Option+D
EDIT	F5	Ctrl+E, Option+E
COPY	Ctrl+C or Option+C	Sends PLATO 0x1B
ANS	Ctrl+A or Ctrl+/,	Option+A
TERM	Ctrl+T	Option+T
SQUARE	Ctrl+Q	Option+Q
ACCESS	Shift+Ctrl+Q	Shift+Option+Q
SUPER	Up Arrow or Page Up	Ctrl+P
SHIFT-SUPER	Shift+Up / Shift+PgUp	Shift+Ctrl+P
SUB	Down Arrow or Page Down	Ctrl+Y
SHIFT-SUB	Shift+Down / Shift+PgDn	Shift+Ctrl+Y
MICRO	Ctrl+M or Option+M	Sends PLATO 0x14
FONT	Ctrl+F or Option+F	Sends PLATO 0x34
TAB / STEP	Tab or Right Arrow	ASCII 0x09 with modulo wrap
ASSIGN	Option+Left Arrow	Avoids Mission Control conflict
MULTIPLY	Ctrl+X or Option+X	PLATO multiplication symbol
DIVIDE	Ctrl+G or Option+G	PLATO division symbol

5.1 Fine Grained Touch and coarse touch

Clicks inside the active terminal square are converted to 512 x 512 logical coordinates. PlatoLives sends the six-byte FGT sequence ESC 0x1F <x_lo> <x_hi> <y_lo> <y_hi>, followed by the legacy 16 x 16 keycode 0x100 | (touch_x << 4) | touch_y. Sending both representations maintains compatibility with coarse lessons and precision pointing software.

5.2 Touchscreen curvature inversion

When Cylinder or Barrel distortion is active, mouseDown: first removes letterbox or pillarbox offsets and normalizes the physical point to u,v in [-1,1]. It then solves the inverse of the selected forward warp. Cylindrical mode inverts only x while leaving y exactly flat. Barrel mode inverts both axes using a short fixed-point iteration. The resulting normalized point is clamped to the optical aperture and converted to the exact logical PLATO cell:

```
x_logical = clamp(floor((x_u + 1) * 256), 0, 511)
y_logical = clamp(floor((y_u + 1) * 256), 0, 511)
```

This inverse mapping is essential: the visual bend affects only presentation and must never alter the host-visible input coordinate contract. Split Monochrome disables distortion and therefore uses direct rectilinear mapping.

5.3 Clipboard integration

Cmd+V uses a GCD queue with 250 ms between printable characters and 500 ms after normalized line breaks, preventing host input overruns. Cmd+. cancels the active paste. Cmd+C captures the current display as a high-resolution PNG on the macOS pasteboard.

6. Display Rendering Modes, Distortion, and Persistence

6.1 The five display modes

Mode	Logical source	Optical behavior	Primary use
Crisp Monochrome	1-bit bitboard	Exact 1:1 rectilinear expansion	Protocol and font verification
Real Plasma	1-bit + plasma energy	Orange neon cells, diffusion, bloom, persistence, optional curvature	Authentic PLATO IV presentation
Split Monochrome	1-bit bitboard	Real Plasma left; Crisp right; no curvature	Immediate optical comparison
Crisp Color	32-bit BGRA matrix	Exact rectilinear color expansion	Color protocol verification
Real Color CRT	32-bit BGRA + P22 energy	Beam reconstruction, mask, scanlines, bloom, persistence, curvature	CDC 721 / IST-III color simulation

6.2 Reorganized View menu

```
+-----+
| View                                     |
| MONOCHROME                             |
|   Crisp Monochrome                     |
|   Split Monochrome                     |
| * Real Plasma                           |
|   Real Plasma Persistence               > |
|   Real Plasma Distortion               > |
|   None | Barrel | Cylindrical          |
|-----+
| COLOR                                   |
|   Crisp Color                           |
|   Real Color CRT                       |
|   CRT Beam Profile                     > |
|   Standard | High | Ultra              |
|   CRT Persistence                     > |
|   20 | 200 | 500 | 1000 | 2000 | 5000 ms |
|   CRT Distortion                     > |
|   None | Barrel | Cylindrical          |
|-----+
| Show Keyboard Reference                 Cmd+K |
| Toggle Full Screen                     F11    |
| Show Performance HUD                   F12    |
+-----+
```

6.3 Real Plasma

Real Plasma simulates the Owens-Illinois neon gas-discharge panel on the 2048 x 2048 optical surface. Each logical cell expands to a calibrated 4 x 4 footprint. Local radius-4 and wide radius-12 separable blur stages represent inter-cell glow and glass scattering. Configurable persistence ranges from 100 ms to 5000 ms with $\tau = \text{duration} / 10.0$. Full-screen Real Plasma supports None, Barrel, and Cylindrical distortion. Split Monochrome remains strictly flat.

6.4 Real Color CRT

Real Color CRT reconstructs the 512 x 512 BGRA matrix into a continuous 2048 x 2048 RGBA surface. A three-tap analog beam distributes each source sample across neighboring positions using a Gaussian profile. A chromatic shadow-mask table modulates red, green, and blue triads without reducing total light and with zero blue crosstalk. A 512-line groove function applies soft horizontal scanlines. Two additive blur scales, radius 4 and radius 12, create local halation and smoked-glass bloom.

Selectable beam profiles

Profile	Sigma	Bloom	Scanline groove	Gain
Standard - Authentic 13-inch CRT	1.8	0.32	22%	1.22
High - Soft Glow (default)	2.4	0.38	20%	1.24
Ultra - Vintage Arcade	2.8	0.45	18%	1.26

6.5 Calibrated optical distortion

The PLATO area occupies the central square of a 13-inch 4:3 tube. Cylindrical is the default because many terminals exhibit predominantly horizontal glass curvature: x bends with $k = 0.018$ while y remains perfectly flat, keeping every scanline straight. Barrel applies proportional spherical curvature with $k_x = 0.018$ and $k_y = 0.012$. None performs identity mapping. Samples outside the warped aperture become 0xFF000000 and merge invisibly with the surrounding black frame.

6.6 P22 and neon persistence

Real Plasma and Real Color CRT share the same physical law but maintain separate state and duration menus. Plasma offers 100, 200, 500, 1000, 2000, and 5000 ms. CRT offers 20 ms P22 Fast, 200 ms Warm Glow, 500 ms Medium, 1000 ms Radar, 2000 ms Ultra, and 5000 ms Storage Tube. CRT persistence stores the host-selected base color in `p->logical` and floating radiant energy in `p->energy`. When energy falls below 0.0001 it is clamped to zero, eliminating plateaus, snaps, and permanent burn-in.

6.7 Zero idle CPU

After all persistence cells have reached zero, no tile remains dirty and `terminal->fb.dirty` is cleared. The display scheduler ceases 60 FPS redraw requests and the process reaches 0.0% idle CPU. New network data, input, window exposure, mode change, or profile change wakes presentation immediately.

7. Implemented Features in Version 3.5

Area	Implemented in 3.5
Transport	Non-blocking TCP, TCP_NODELAY, reconnect-safe worker, ring buffer, diagnostic log, IST-III negotiation.
Color protocol	Subtype-16 ESC Y 0x71 response and full 32-bit color stream handling.
Logical display	Parallel 512 x 512 one-bit bitboard and 512 x 512 little-endian BGRA color matrix.
Graphics semantics	Points, Bresenham vectors, blocks, write/erase/rewrite, canonical inverse to bg_color.
Coordinates	Bottom-left logical origin and modulo-512 Backspace/Step toroidal wrap.
Typography	Canonical M0/M1, programmable M2/M3, continuous CDC Appendix G EXT scaling and tracking.
Display modes	Crisp Monochrome, Real Plasma, Split Monochrome, Crisp Color, Real Color CRT.
CRT optics	Three-tap Gaussian beam, pure chromatic mask, scanlines, dual-radius additive halation/bloom.
Beam presets	Standard, High default, and Ultra, each with calibrated sigma, bloom, groove, and gain.
Distortion	None, Cylindrical default, and Barrel for Real Plasma and Real Color CRT; Split forced flat.
Persistence	Neon 100-5000 ms and P22 20-5000 ms with continuous float exponential decay.
Input	PLATO key engine, FGT plus coarse touch, and inverse curved-glass click mapping.
Profiles	Profile-specific plasma and CRT controls with dynamic visibility and menu synchronization.
Windows	Independent multi-window sessions, CALayers, timers, sockets, terminal state, and PMD titles.
Clipboard	Paced paste, cancellation, and high-resolution Copy Screen.
Performance	O(1) blur, boundary tile dilation, native BGRA presentation, zero idle CPU, Mach telemetry.
Automation	Command files for keys, text, waits, modes, screenshots, diagnostics, and controlled quit.
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8. Features Still to Be Implemented

Version 3.5 completes the color, optical geometry, and continuous-persistence work described in this manual. Remaining roadmap items are deliberately outside the available feature set:

Feature	Planned direction
Mouse / trackpad scrolling	Implement scrollWheel: translation through the appropriate extended terminal input protocol.
Numpad 8-way movement	Add configurable numeric-keypad movement for dungeon and maze applications.
Character Window Save/Restore	Implement full CWS behavior for TERM prompt and window workflows.
Generalized Sound Writer audio	Decode GSW packets and synthesize period-appropriate musical and tone output.
Calibration export/import	Package optical presets as portable, versioned profile records.
Accessibility controls	Add user-selectable high-contrast overlays without changing host-visible framebuffer semantics.
Windows 11 port	Retain libplato and port presentation to native Win32 and Direct3D.
Automated conformance corpus	Expand screenshot and protocol fixtures across classic lessons and games.

9. Diagnostic Logging

9.1 Enabling the log

Choose Tools -> Enable Diagnostic Log. The item changes to Disable Diagnostic Log while active. Transport traffic is written to PlatoLives.log in the application directory.

9.2 Captured information

- Session headers: timestamp, PID, direction, byte count, and stream offsets.
- Raw dumps: hexadecimal and printable ASCII before decoding.
- Protocol events: normalized Telnet pairs, escape sequences, terminal modes, color changes, and outgoing keys.
- Optical state: renderer changes, distortion selection, persistence duration, beam profile, dirty-tile transitions, and quiescence entry/exit.
- Metadata: PMD user, group, and slot when supplied by the host.

```
[22:04:03.709] [RX] packet=12 bytes=28 stream=[0x1A40..0x1A5B]
RAW 00001A40: 1B 59 70 40 40 1B 63 40 40 1F 5A ...
EVENT 00001A40 ESC Y subtype=16 response=0x71 COLOR-CAPABLE
EVENT renderer=real-color-crt beam=high persist=20ms distort=cyl
PERF dirtyTiles=0 fb.dirty=0 state=QUIESCENT
```

10. Command File Automation

The command-file runner supports repeatable login, protocol regression, optical comparison, and screenshot generation:

```
PlatoLives --test-script ~/Developer/PlatoLives/test-platolives.txt
```

Command	Behavior
send <text>	Send raw ASCII text.
send-env <VAR>	Send an environment value with redacted logging.
key <NAME>	Send a named PLATO key.
wait <duration>	Wait asynchronously, for example 2s or 500ms.
renderer <mode>	Select crisp-mono, real-plasma, split-mono, crisp-color, or real-color-crt.
beam <profile>	Select standard, high, or ultra CRT beam profile.
persistence <duration>	Select the duration valid for the active analog renderer.
distortion <mode>	Select none, barrel, or cylindrical where supported.
screenshot <file>	Write the internal 2048 x 2048 display to PNG.
diagnostic-log <on off>	Toggle transport and renderer logging.
quit	Terminate cleanly after outstanding work completes.

11. Performance Strategy

11.1 O(1) separable sliding-window blur

The radius-4 local diffusion and radius-12 wide bloom passes use running sums. Each destination sample costs one entering add, one leaving subtract, and one scale, independent of radius. Horizontal passes stream contiguous cache lines. Vertical passes use a compact column accumulator rather than repeatedly walking strided neighborhoods.

$$S(x) = S(x-1) + \text{source}[x+R] - \text{source}[x-1-R]$$

$$\text{dest}[x] = S(x) / (2R+1)$$

11.2 Boundary-aware tile dilation

The logical display is divided into 16 x 16 tiles. Optical work expands from a dirty tile only when active or decaying samples lie close enough to an edge for a radius-12 kernel to affect its neighbor. The dilation margin is three logical pixels, equal to twelve supersampled pixels. This bounds bloom correctly while avoiding unconditional full-frame recomputation.

11.3 Native BGRA and Metal-aligned presentation

The 2048 x 2048 output is little-endian BGRA, matching kCGBitmapByteOrder32Little and MTLPixelFormatBGRA8Unorm. A wantsLayer-backed Cocoa view presents through CALayer in sRGB. Matching memory and texture formats eliminates per-frame channel swizzles and unnecessary ColorSync conversions. The same aligned surface serves monochrome and color modes.

11.4 Persistence scheduling and zero idle CPU

A tile remains scheduled while it contains active phosphor or neon energy, a logical update, or optical state invalidation. Once p->energy is zero everywhere, dirtyTiles equals zero, and terminal->fb.dirty is cleared, no periodic frame is requested. This is genuine quiescence rather than rendering a static frame sixty times per second. Input and network callbacks atomically re-arm the display scheduler.

11.5 Mach thread telemetry

F12 displays measured FPS, partial/full work, user/slot metadata, dirty-tile count, renderer parameters, and process CPU. CPU time is obtained from task_threads() and THREAD_BASIC_INFO, accumulated across live threads, normalized by elapsed wall time, and adjusted for the platform Mach tick scale. Sleeping network and presentation workers therefore appear correctly as 0.0% in a static, fully decayed session.

11.6 Expected operational profile

State	Expected behavior
Static crisp display	No renderer work until exposure, input, or network change.
Static analog display after decay	dirtyTiles = 0, fb.dirty = 0, presentation scheduler stopped, 0.0% CPU.
Active vector animation	Partial tiles dominate; 60 FPS target maintained.
Very large color update or mode change	Full pass permitted, then returns to partial work or quiescence.
Window resize / full screen	Optical surface is reprojected while preserving the 1:1 logical square.

12. Troubleshooting

Symptom	Probable cause	Recommended action
** CONNECTION FAILED **	Host unavailable, wrong port, or offline network.	Verify the profile and reconnect. The message is drawn in the terminal without blocking dialogs.
Terminal type queried	Handshake desynchronized.	Disconnect and reconnect; ESC Y negotiation restarts automatically.
Color lesson remains monochrome	Crisp Monochrome or non-color host path active.	Select Crisp Color or Real Color CRT and verify subtype-16 negotiation in the log.
Curved picture but inaccurate clicks	Old profile or stale geometry state.	Re-select the distortion mode or reconnect; mouseDown: then applies inverse warp.
Scanlines bend vertically	Barrel distortion selected.	Choose Cylindrical to keep the y axis and scanlines strictly flat.
Split view appears flat	Expected behavior.	Split Monochrome deliberately disables curvature for rectilinear comparison.
Long trail remains too long	Persistence duration is high.	Select P22 Fast 20 ms for CRT or Authentic 100 ms for plasma.
CPU remains active on static screen	Energy still decaying or diagnostic exposure is active.	Wait for the selected decay interval; inspect dirtyTiles and fb.dirty in F12/log.
Halved glyph at left edge	Outdated build without modulo wrap.	Use Version 3.5 and retest Backspace/Step at x=0.
Password box is not cleared	Non-canonical inverse behavior.	Use Version 3.5; mode 3 fills with bg_color.
Mission Control intercepts Assign	macOS shortcut conflict.	Use Option+Left Arrow.
Paste drops characters	Host input buffer overrun.	Use Cmd+V paced paste and avoid unpaced external paste.
Greek or symbol characters	M1 is active.	Press FONT to return to M0 if the host lesson expects it.

Appendix A. Real Plasma Rendering: Technical Design

A.1 Historical and logical model

The original PLATO IV display was an AC gas-discharge neon plasma panel with inherent bistable memory. PlatoLives separates its logical state from optical presentation. Protocol decoding, vectors, blocks, and font blits operate on the 512 x 512 bitboard; the optical stage expands the result to a continuous 2048 x 2048 surface.

A.2 Four-by-four cell footprint

```
[ 0.65  0.82  0.82  0.65 ]
[ 0.82  1.00  1.00  0.82 ]
[ 0.82  1.00  1.00  0.82 ]
[ 0.65  0.82  0.82  0.65 ]
```

A stable per-cell gain in [0.98,1.02] reproduces fixed panel variation without temporal noise.

A.3 Neon persistence

$$E(t + dt) = E(t) \exp(-dt / \tau), \quad \tau = T_{\text{decay}} / 10$$

When $E < 0.0001$, energy is clamped to zero. The tile then leaves the active set unless another logical or optical dependency remains.

A.4 Dual separable blur

Local radius $R = 4$ models inter-cell diffusion. Wide radius $R = 12$ models glass scattering. Each two-dimensional box blur is separated into horizontal and vertical passes using the running-sum equations in Section 11.1.

A.5 Density fusion and orange response

$$n = \text{clamp}((\text{local_glow} - 0.25) / 0.40, 0, 1)$$

$$\text{fused} = \text{core} + 0.82 n (1 - \text{core}), \quad \text{for } \text{core} > 0.05$$

The fused core, local glow, and wide glow are combined with a restrained orange-biased response and written directly to the native BGRA presentation surface.

Appendix B. Real Color CRT and Optical Geometry: Mathematical Design

B.1 Three-tap Gaussian electron beam

For a logical sample centered at x_0 , the beam contributes to three adjacent supersample positions i in $\{-1, 0, +1\}$. With selectable sigma in $\{1.8, 2.4, 2.8\}$:

```
g_i = exp(-(i^2) / (2 sigma^2))
w_i = g_i / (g_-1 + g_0 + g_+1)
C_beam(x0+i, y) += gain * w_i * C_logical(x0, y)
```

The same normalized kernel is applied consistently across the two-dimensional reconstruction so that line energy is conserved before profile gain, scanline modulation, mask modulation, and additive bloom.

B.2 Pure chromatic shadow-mask table

The repeating four-position mask is indexed by supersample x modulo four. The implementation uses channel-selective boosts with no blue leakage into red or green sites and a neutral wire position. A normalized conceptual table is:

```
mask_tbl[4][3] = {
    {R_boost, 1,      0},      // sx = 0, red triad
    {1,      G_boost, 0},      // sx = 1, green triad
    {0,      0,      B_boost}, // sx = 2, blue triad
    {1,      1,      1}       // sx = 3, matrix wire
}
```

Profile calibration renormalizes aggregate luminous energy, so the mask does not impose artificial light loss. The blue channel has zero crosstalk by construction at non-blue sites.

B.3 Soft analog scanlines

For supersample row y_s , logical cathode phase $\phi = 2 \pi (y_s \bmod 4) / 4$. A smooth groove modulation with depth d is applied:

```
m_scan(y_s) = 1 - d * (1 - cos(phi)) / 2
C_scan = m_scan * C_mask,    d in {0.22, 0.20, 0.18}
```

Cylindrical distortion leaves y unchanged, so these scanlines remain exactly straight.

B.4 Calibrated forward distortion

Let x, y be normalized undistorted coordinates in $[-1, 1]$. The default cylindrical projection uses $k = 0.018$ only on x :

```
x_d = x (1 + k_cyl_x x^2),    k_cyl_x = 0.018
y_d = y
```

The barrel projection uses the physical 4:3 tube proportions through separate coefficients:

```
r2 = x^2 + y^2
x_d = x (1 + k_bar_x r2),    k_bar_x = 0.018
y_d = y (1 + k_bar_y r2),    k_bar_y = 0.012
```

None is the identity mapping $x_d=x, y_d=y$. Samples outside the valid curved aperture are assigned 0xFF000000.

B.5 Inverse touch mapping

Cylindrical inversion solves $x_d = x(1+kx^2)$ for x with Newton iteration while $y=y_d$:

```
x_(n+1) = x_n - (x_n + k x_n^3 - x_d) / (1 + 3 k x_n^2)
```

Barrel inversion starts with $x_0=x_d, y_0=y_d$ and repeatedly removes the radial scale:

```
r_n^2 = x_n^2 + y_n^2
x_(n+1) = x_d / (1 + kx r_n^2)
y_(n+1) = y_d / (1 + ky r_n^2)
```

Because k is small, a few bounded iterations converge monotonically over the aperture.

B.6 Continuous P22 color-phosphor decay

For each color cell, L is the host-selected base color in $p \rightarrow \text{logical}$ and E is scalar luminous energy in $p \rightarrow \text{energy}$:

```
E(t + dt) = E(t) exp(-dt / tau),    tau = T_CRT / 10
C_emit(t) = L * E(t)
if E(t) < 0.0001, then E(t) = 0 and C_emit = black
```

The threshold corresponds to 0.01% light. It creates neither a visible plateau nor a permanent residual, and it provides the exact terminal condition required for zero-idle-CPU quiescence.

B.7 Additive halation and profile composition

```
C_out = clamp(gain * C_scan + bloom * (B_4(C_scan) + B_12(C_scan)), 0, 1)
```

B_4 and B_{12} are separable box blurs of radii four and twelve. Bloom is chromatic and purely additive, representing scattering in smoked front glass rather than desaturation toward white.

Appendix C. Historical and Technical References

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9. Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International Public License, SPDX CC-BY-NC-SA-4.0.

Version 3.5 release summary

Version 3.5 unifies historical terminal correctness with restrained physical display simulation. The protocol core now supports full-color CDC 721 / IST-III streams, canonical inverse drawing, continuous EXT character scaling, and modulo-512 cursor motion. The presentation layer adds a calibrated Real Color CRT, cylindrical and barrel tube geometry, exact inverse touch mapping, continuous P22 decay, profile-aware analog controls, and true static-screen quiescence. These changes preserve the 512 x 512 host contract while allowing the modern macOS implementation to reproduce both the digital data and the optical character of the original display families.